

Getting Started with the Smart Fish

The Smart Fish Kit is an educational product tailored for primary and secondary school students, as well as electronic DIY enthusiasts, to deliver an engaging hands-on project experience.

Below is a step-by-step guide to getting started with the Smart Fish :

Installing the Control Software

First, we need to set up the control software for the Smart Fish Kit.

The Smart Fish Kit is controlled via Bluetooth protocol, so you will need to install a Bluetooth command utility (e.g., BLE Debug Assistant/Bluetooth Debugger) on your computer or mobile phone.

Let's take the mobile version of Bluetooth Debugger as an example. (The installation package for the software is available in the current folder: [Bluetooth Debugger](#))

Once installed, the app icon will appear on your mobile phone screen.



Of course, any other Bluetooth debugging software is also compatible.

Powering On the Device

The Smart Fish Kit uses a magnetic control switch. To power it on:

1. Prepare a piece of adhesive tape and a magnet.
2. Attach the magnet to the adhesive tape.



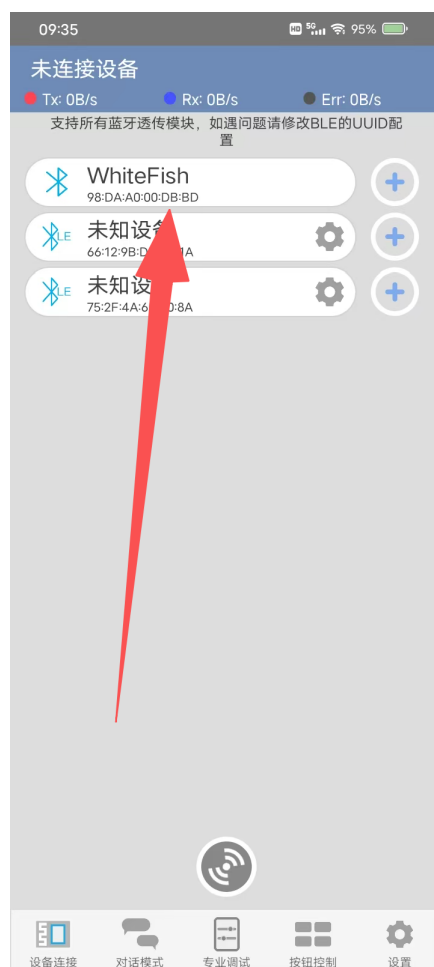
3. Stick the tape (with the magnet) to the top of the Smart Fish Kit, as shown below.



A "click" sound (from the relay activation) confirms that the Smart Fish has been powered on successfully.

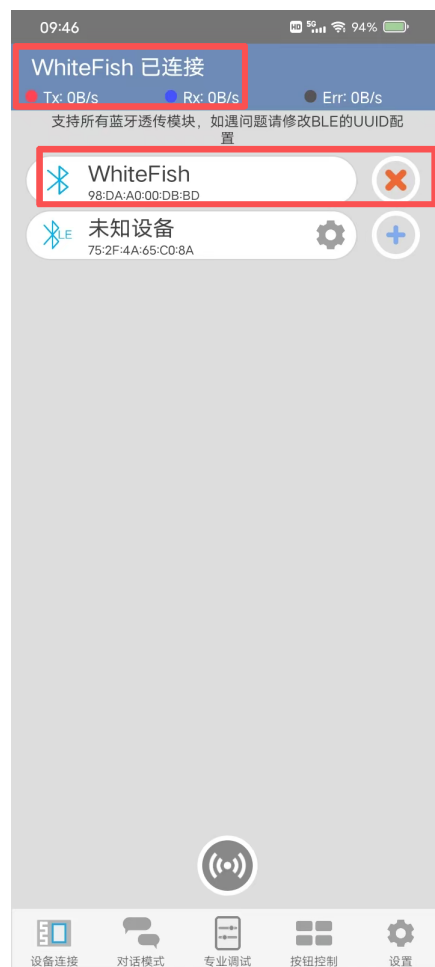
Connecting to the Smart Fish Kit

1. Launch the installed Bluetooth software.
2. Scan for Bluetooth devices and connect to the Smart Fish Kit's Bluetooth signal.



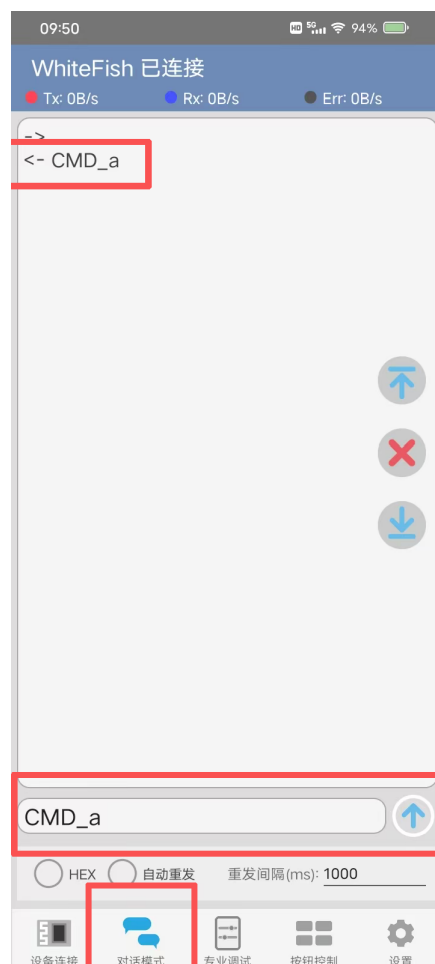
If no devices are detected by the Bluetooth Debugger, check that your phone's location and Bluetooth services are enabled.

After a successful connection, the software interface will look like this:



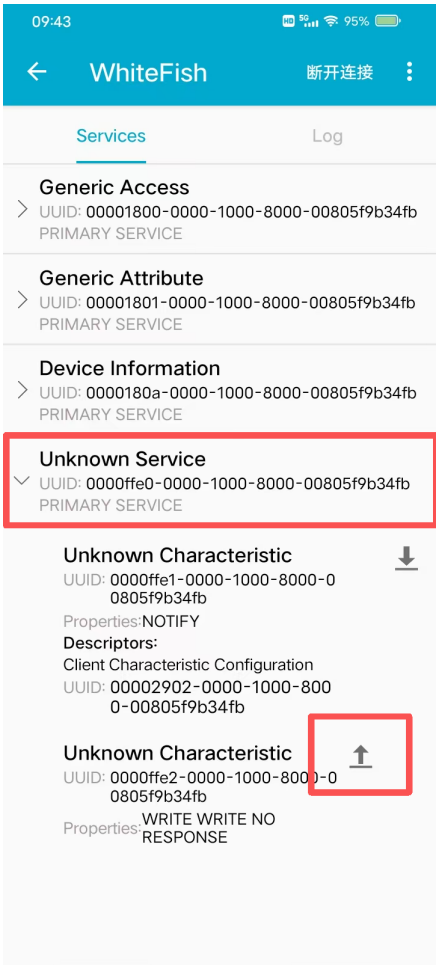
A PIN code may be required for the first connection. Enter 0000 or 1234 if prompted.

3. Once connected, enable chat mode and send the command `CMD_a`.



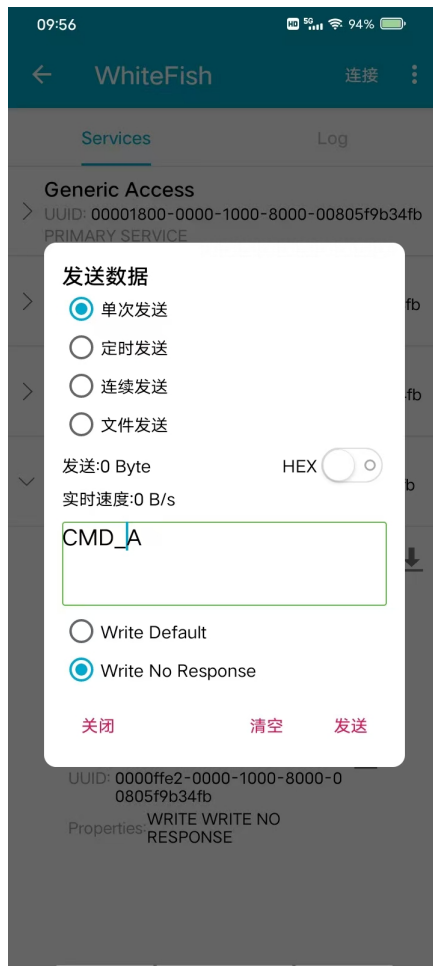
A slight wag of the fish's tail confirms that the Smart Fish Kit is functioning properly.

For other professional Bluetooth debugging software, the post-connection interface may look like this:



In this case:

1. Select `Unknown Service`.
2. Choose the "Upload Service" option.
3. Send the corresponding commands here.



Additional Commands

Once the Smart Fish Kit is operational, you can try sending additional commands. Basic commands are listed below:

Command	Function Description
CMD_S	Return tail to center position
CMD_R	Continuous tail wag to the right
CMD_L	Continuous tail wag to the left
CMD_a	±15° wag (low speed)
CMD_A	±15° wag (high speed)
CMD_b	±30° wag (low speed)
CMD_B	±30° wag (high speed)
CMD_c	±45° wag (low speed)
CMD_C	±45° wag (high speed)

For more custom advanced commands, refer to the document in the folder: [Smart Fish Control Command Guide](#)

Customizing Control Buttons

Repeatedly editing command text can be cumbersome during operation. To simplify this:

1. Open the following interface in the Bluetooth Debugger:

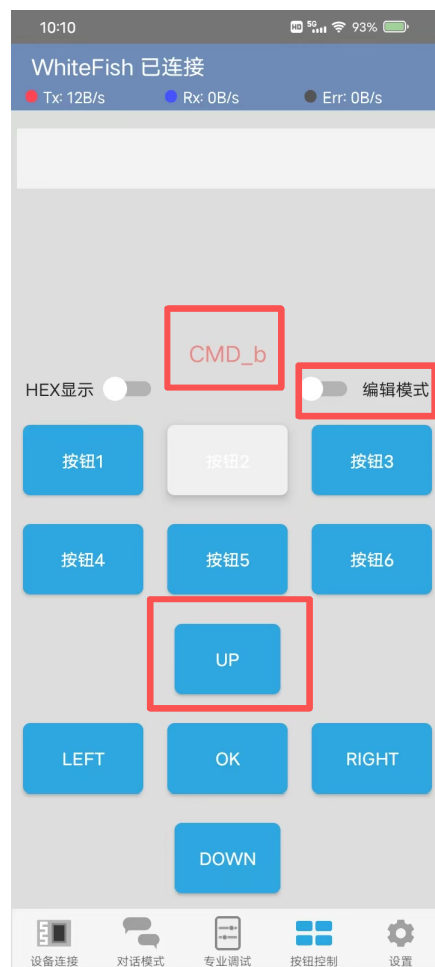


2. Enable edit mode.
3. Select a button to customize the command it sends.



4. Edit the button name and the data sent when the button is pressed or released (only one of the "press" or "release" event data fields needs to be configured).

IMPORTANT: Do NOT enable HEX transmission!!!



Notes for Underwater Use

If the Smart Fish Kit is to be used underwater, additional waterproofing measures are mandatory, including but not limited to:

- Add and adjust counterweights to ensure the kit floats semi-submerged and balanced on the water surface.
- Encase internal electronic components in a waterproof sealable bag for protection.
- Wrap Teflon tape around internal knob connections to enhance water resistance.
- Apply waterproof tape to external joints of the kit to ensure full system waterproofing.